



Islamic University of Technology
Department of Computer Science and Engineering

Lab 3: Graph Traversal

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

Contents

Task A. Shortest Path in Sundarbans	2
Task Statement	2
Input Format	2
Output Format	2
Examples	2
Notes	3
Task B. Course Prerequisite Order	4
Task Statement	4
Input Format	4
Output Format	4
Examples	4
Notes	5

Task A. Shortest Path in Sundarbans

Time Limit: 1 second | Memory Limit: 256 MB

Motu is on an environment protection mission deep in Sundarbans. The region has n workstations connected by m bidirectional paths, where each path takes exactly one unit of time to traverse.

Motu is currently at workstation a and must reach workstation b for an urgent briefing. Help him find the minimum travel time to get there.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains two integers n and m ($2 \leq n \leq 2 \times 10^5$, $1 \leq m \leq 5 \times 10^5$) — the number of workstations and paths.
- The next m lines each contain two integers u and v ($1 \leq u, v \leq n$, $u \neq v$) — a bidirectional path between workstations u and v .
- The last line contains two integers a and b ($1 \leq a, b \leq n$, $a \neq b$) — Motu’s current station and the meetup station.

It is guaranteed that the graph is connected. The sum of n over all test cases does not exceed 2×10^5 , and the sum of m does not exceed 5×10^5 .

Output Format

For each test case, print a single integer — the minimum travel time from a to b .

Examples

Sample Input	Sample Output
2	1
4 5	2
1 2	
1 3	
2 3	
2 4	
3 4	
1 3	
4 4	
1 2	
2 3	
3 4	
2 4	
1 4	

Notes

In the first test case, the direct path $1 \rightarrow 3$ takes 1 unit of time.

In the second test case, the shortest path from 1 to 4 is $1 \rightarrow 2 \rightarrow 4$ or $1 \rightarrow 3 \rightarrow 4$, both taking 2 units of time.

Task B. Course Prerequisite Order

Time Limit: 1 second | Memory Limit: 256 MB

A university offers n courses numbered 1 to n . Some courses have prerequisites: to take course v , you must have already completed course u (denoted as a directed edge $u \rightarrow v$).

A student wants to take all n courses. Help the student find a valid order to take the courses such that all prerequisites are satisfied before taking each course.

If no valid order exists (i.e., there is a circular dependency among courses), print -1 .

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains two integers n and m ($1 \leq n \leq 2 \times 10^5$, $0 \leq m \leq 5 \times 10^5$) — the number of courses and the number of prerequisite relations.
- The next m lines each contain two integers u and v ($1 \leq u, v \leq n$, $u \neq v$) — meaning course u must be completed before course v .

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 , and the sum of m does not exceed 5×10^5 .

Output Format

For each test case:

- If a valid order exists, print n space-separated integers — a valid topological ordering of the courses.
- Otherwise, print -1 .

If multiple valid orderings exist, print any one of them.

Examples

Sample Input	Sample Output
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3	1 2 3 4
4 4	-1
1 2	1 2 4 3 5
1 3	
2 4	
3 4	
3 3	
1 2	
2 3	
3 1	
5 3	
1 3	
2 3	
4 5	

Notes

In the first test case, course 1 must come before 2 and 3, and both must come before 4. The order 1 2 3 4 or 1 3 2 4 are both valid.

In the second test case, courses 1, 2, 3 form a cycle ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$), so no valid ordering exists.

In the third test case, courses 1 and 2 have no prerequisites and can come in any relative order. Course 3 requires both 1 and 2. Courses 4 and 5 are independent except that 4 must precede 5.

Hint: DFS-based topological sort. If the resulting order does not include all n nodes, a cycle exists.

Marks Distribution

Task	Marks
Task A	50%
Task B	50%

Practice Problems

Problem	Links
Counting Rooms	CSES: https://cses.fi/problemset/task/1192
Labyrinth	CSES: https://cses.fi/problemset/task/1193
Building Roads	CSES: https://cses.fi/problemset/task/1666
Learning Languages	Codeforces: https://codeforces.com/contest/277/problem/A Vjudge: https://vjudge.net/problem/CodeForces-277A
Kefa and Park	Codeforces: https://codeforces.com/contest/580/problem/C Vjudge: https://vjudge.net/problem/CodeForces-580C